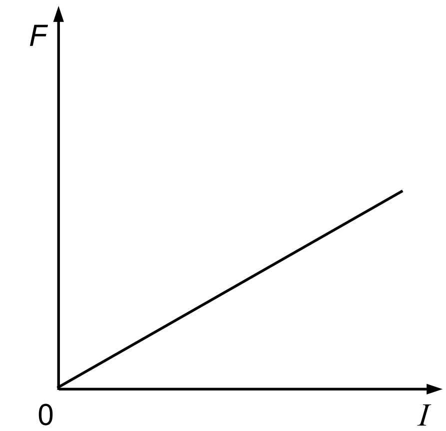
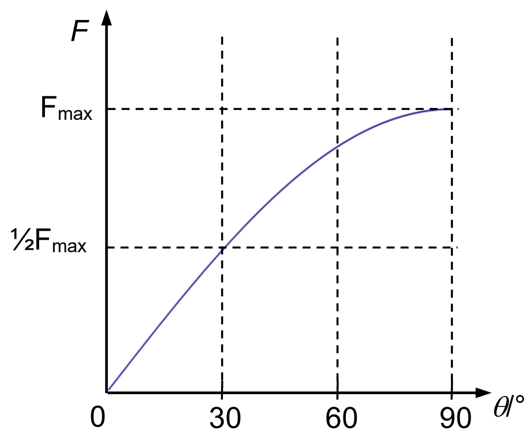
6ai	$F = B_{\perp}Il = BIl \sin\theta$ $F \propto I$ for B, l and θ constant straight line with positive gradient through origin 
6aii	$F \propto \sin\theta$ for B, I and l constant Curve with decreasing gradient from 0° to 90°. Max force at θ = 90° and zero force at θ = 0°. sine curve with zero gradient θ = 90° $\frac{1}{2}F_{max}$ at θ = 30° 
6bi	There is a (magnetic) <u>force</u> acting on electron due to magnetic field. By applying Fleming's Left Hand Rule, this force is directed towards QR. Hence, electron moves towards QR.
6bii	Initially, electrons accumulate on side QR. This sets up an electric field from side PS to side QR. After some time, the electric force acting on subsequent electrons is equal and opposite to the magnetic force acting on these electrons. Hence, subsequent electrons travel undeflected.

7ai	Power dissipated in the resistor is due to the non-zero <u>root-mean-square</u> current.